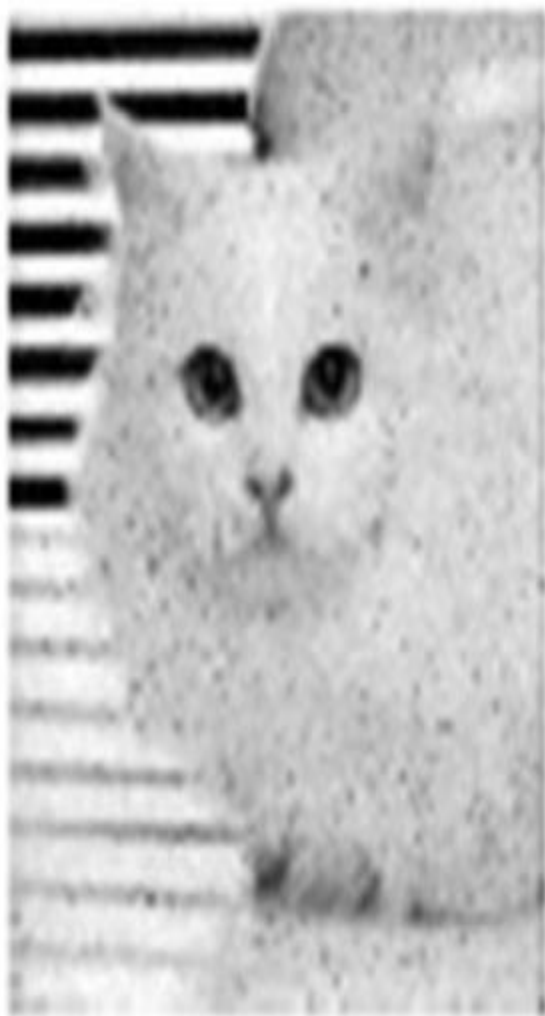


Image is NOT Perfect Sometimes



After Image Enhancement





Why do we need Image Enhancement?

- The goal of image enhancement is to process a digital image and make it more “suitable” than the original image for a specific application
- Images may suffer from the following degradations:
 - ▣ Poor contrast due to poor illumination or finite sensitivity of the imaging device
 - ▣ Electronic sensor noise or atmospheric disturbances leading to broad band noise
 - ▣ Aliasing effects due to inadequate sampling
 - ▣ Finite aperture effects or motion

Example •

A method that is quite useful for enhancing X-Ray images •
may not be the best approach for enhancing infrared images.

Image Enhancement

Spatial Domain

Frequency Domain

Intensity Transformation
(Point Processing)

Spatial Filtering
(Mask Processing)

Linear

Non Linear

Piecewise
Linear

Histogram

Linear
Filter

Non-Linear
Filter

→ Image
Negative

→ Logarithmic
→ Power-Law

→ Contrast Stretching
→ Gray-Level Slicing
→ Bit-Plane Slicing

→ Average
→ Weighted Average
→ Derivative
(Laplacian)

→ Order
Statistics

Median

Minimum

Maximum

Image Enhancement

- There are Two broad categories of Image Enhancement Approaches
 - **Spatial domain methods**
 - Based on direct manipulation of pixels in an image
 - **Frequency domain methods**
 - Based on modifying the Fourier transform of an image

Spatial Domain

- **Intensity Transformations (Point processing)**

- Operate on single pixels of an image

- e.g., image averaging; logic operation; contrast stretching ...

- **Spatial Filtering (Mask processing)**

- Working in a neighbourhood of every pixel in an image

- e.g., blurring, median

Good Images

- For human visual
 - ▣ The visual evaluation of image quality is a highly subjective process.
 - ▣ It is hard to standardize the definition of a good image.
- For machine perception
 - ▣ The evaluation task is easier.
 - ▣ A good image is one which gives the best machine recognition results.
- A certain amount of trial and error usually is required before a particular image enhancement approach is selected.

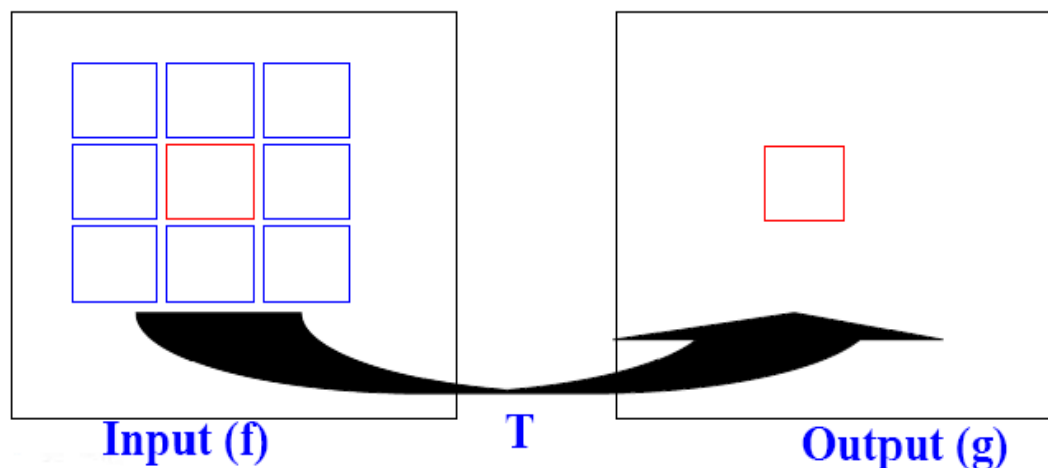
Spatial Domain

- Spatial domain can be denoted by:

$$g(x,y) = T[f(x,y)]$$

where

- $f(x,y)$ is the input image
- $g(x,y)$ is the processed image
- T is an operator on f defined over some neighborhood of (x,y)



Intensity Transformation

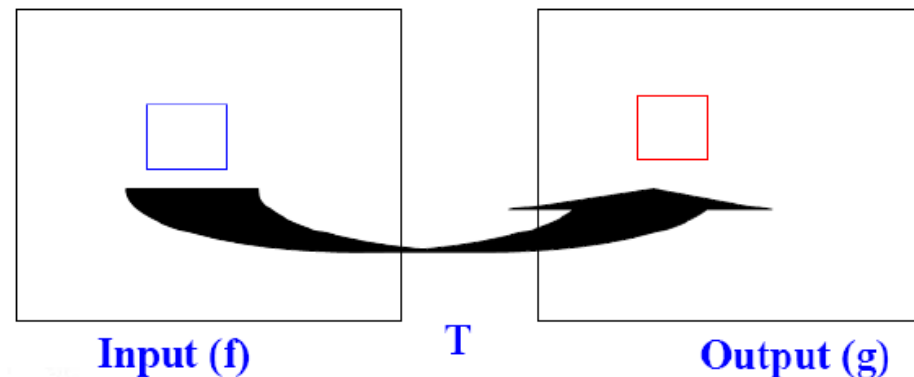
- **Simplest T** : operates on neighborhood of 1×1
- Then, it becomes **intensity transformation**
- g depends on only the value of f at (x,y)
- T = gray level (or intensity or mapping) transformation function

$$s = T(r)$$

- Where

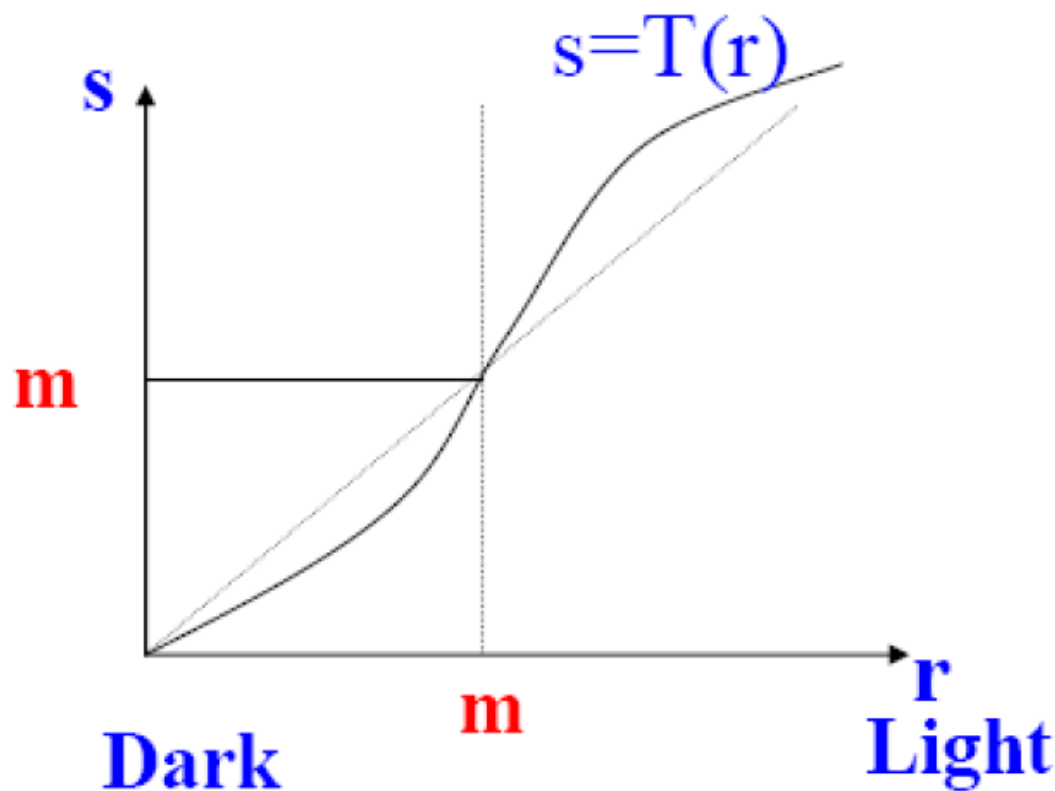
▣ r = gray level of $f(x,y)$

▣ s = gray level of $g(x,y)$



Intensity Transformation ...

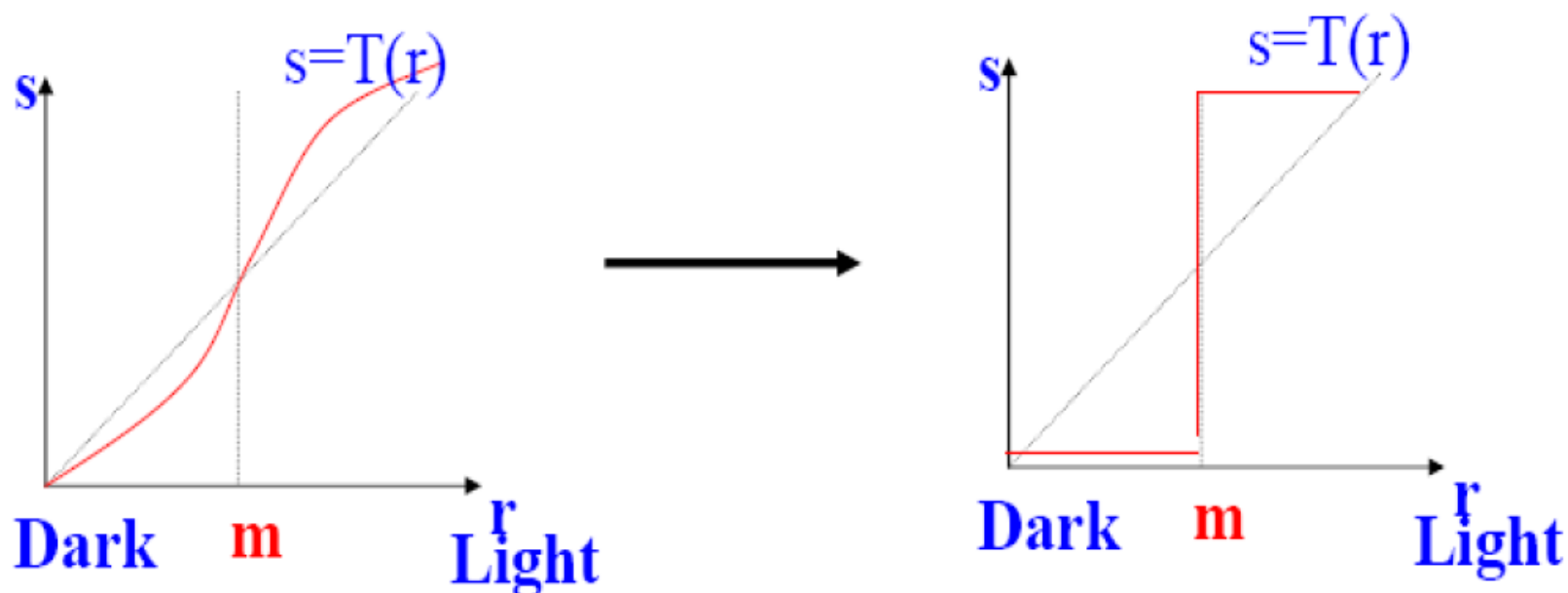
- The gray level below **m** are darkened and the levels above **m** are brightened.



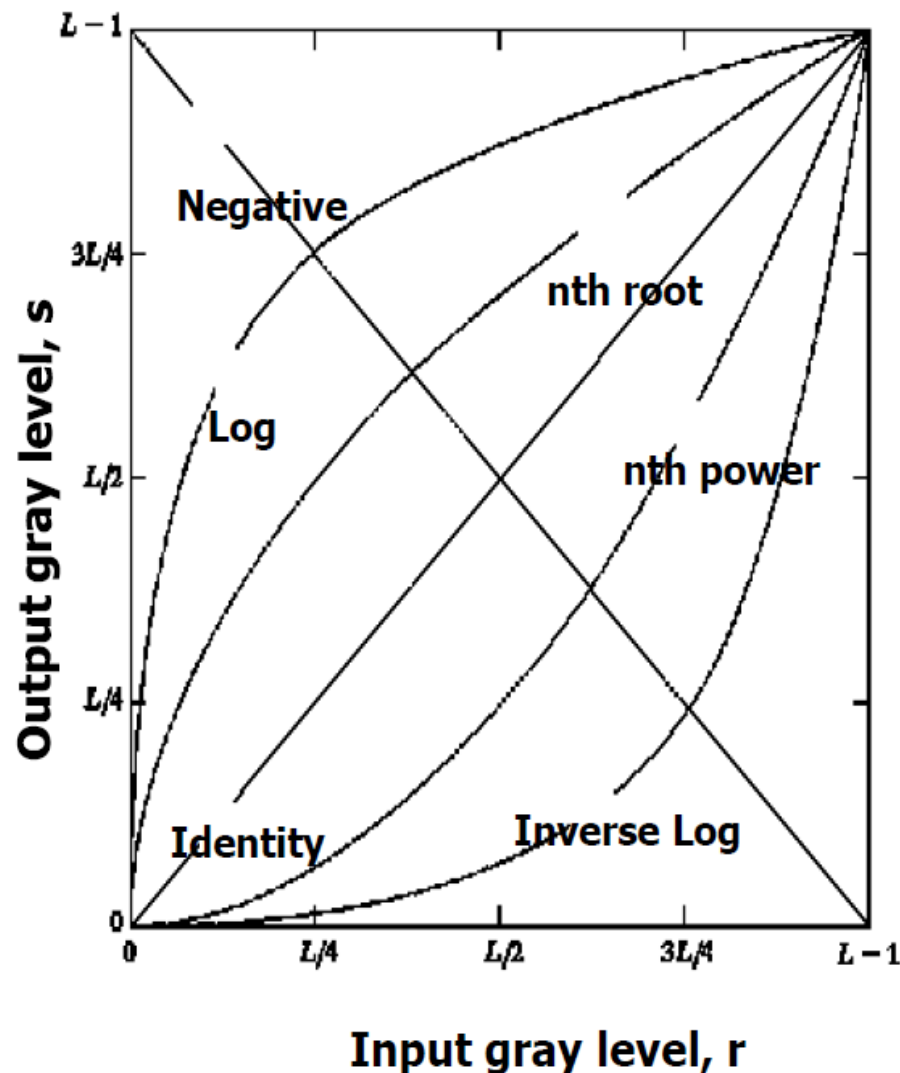
Intensity Transformation ...

- **Limiting case:** Produces a binary image (two level) from input image and the function is known as Thresholding function

$$g(x,y) = \begin{cases} L & \text{if } f(x,y) > m, \\ 0 & \text{otherwise} \end{cases}$$



Basic Intensity Transformation Functions



Linear function

- ▣ Negative and identity transformations

Logarithm function

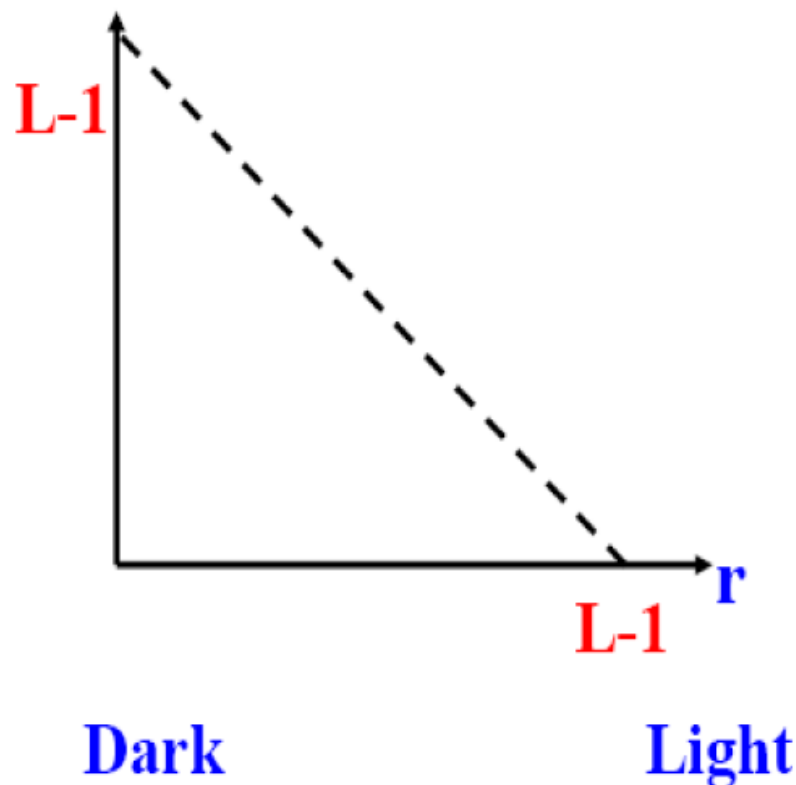
- ▣ Log and inverse-log transformation

Power-law function

- ▣ n^{th} power and n^{th} root transformations

Identity function: Output intensities are identical to input intensities.

Image Negatives



- An image with gray level in the range $[0, L-1]$ where $L = 2^n$; $n = 1, 2, \dots$
- Negative transformation :
$$s = L - 1 - r$$
- Reversing the intensity levels of an image.
- Suitable for enhancing white or gray detail embedded in dark regions of an image, especially when the black area is dominant in size.



Image Negatives

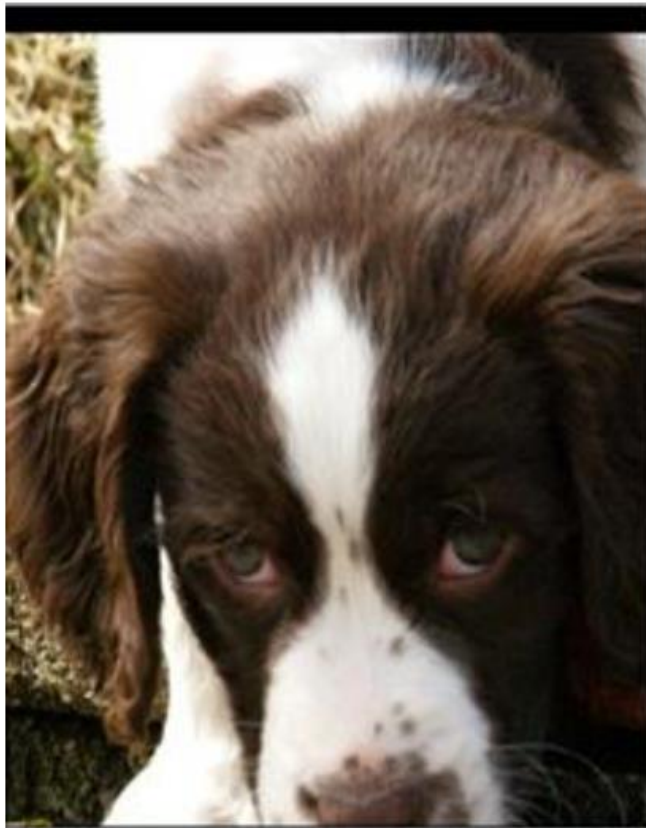


Image Negatives: Example



Image Negatives

Image (r)



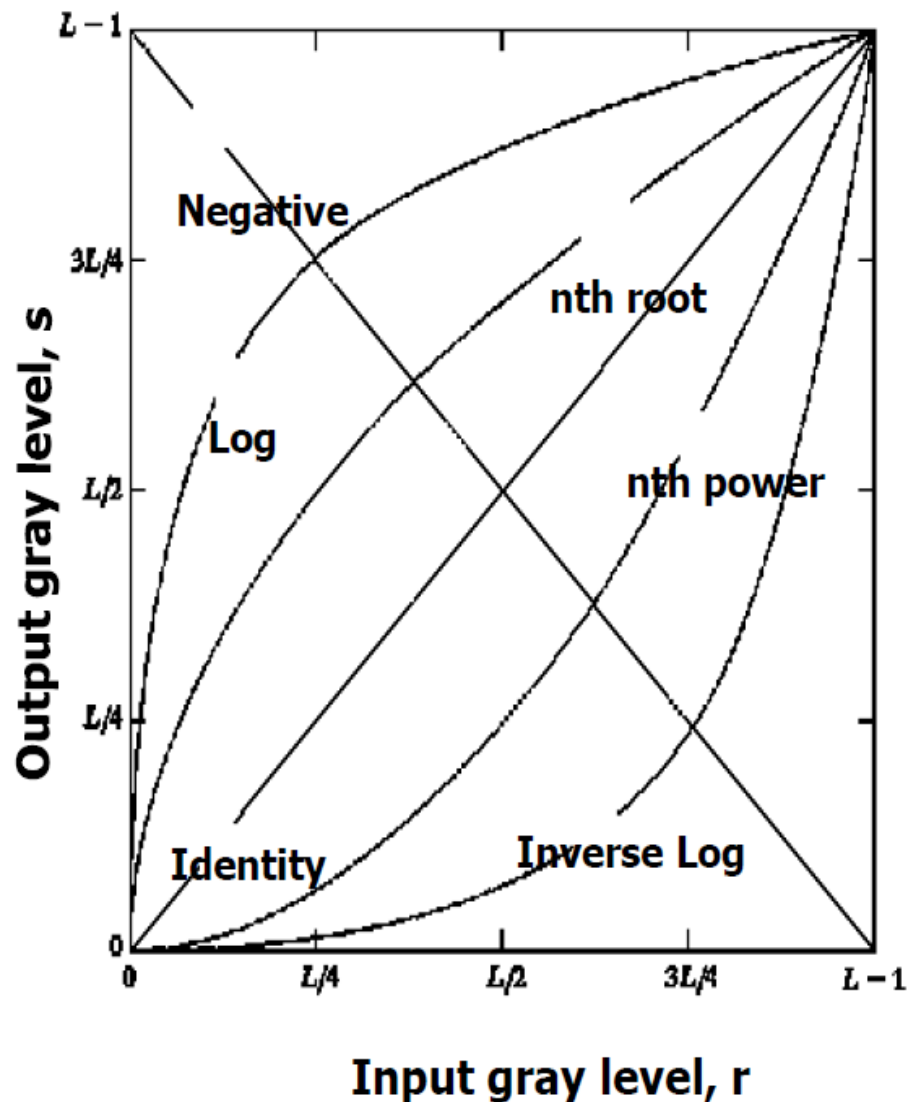
Image (s) after applying T (negative)



Advantages of negative :

- ✓ Produces an equivalent of a photographic negative.
- ✓ Enhances white or gray detail embedded in dark regions.

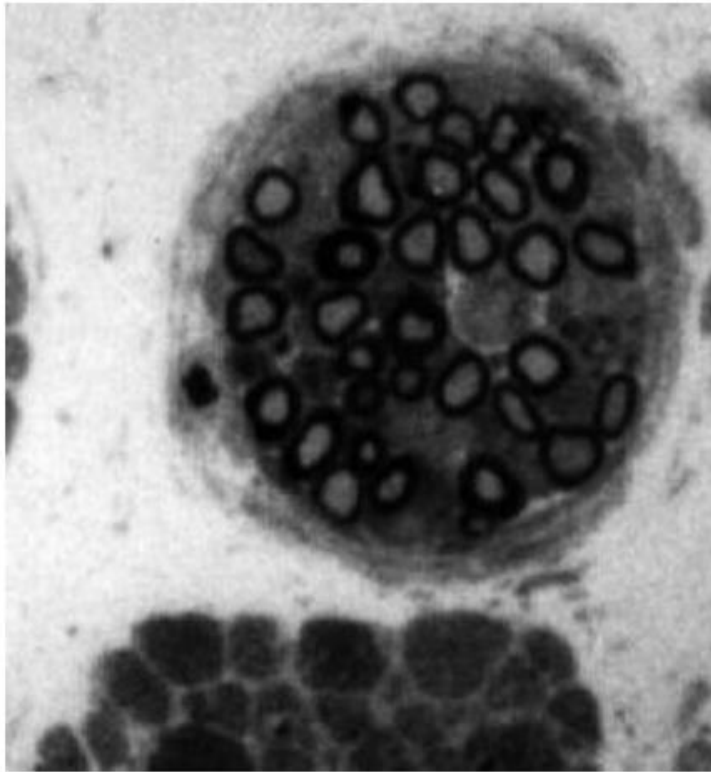
Log Transformations



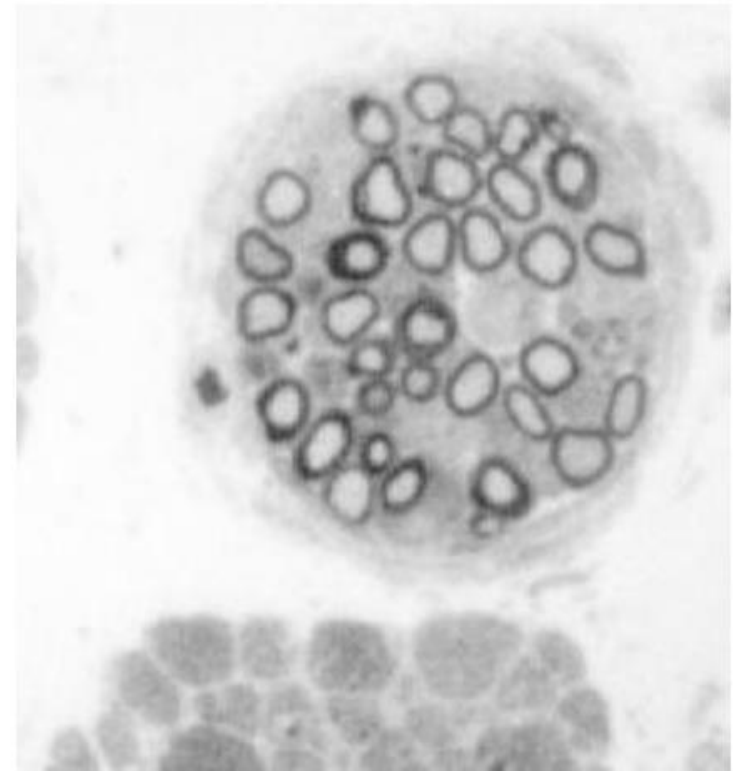
$$s = c \log (1+r)$$

- c is a constant and $r \geq 0$
- Log curve maps a narrow range of low gray-level values in the input image into a wider range of output levels.
- Used to expand the values of dark pixels in an image while compressing the higher-level values.

Log Transformations: Example

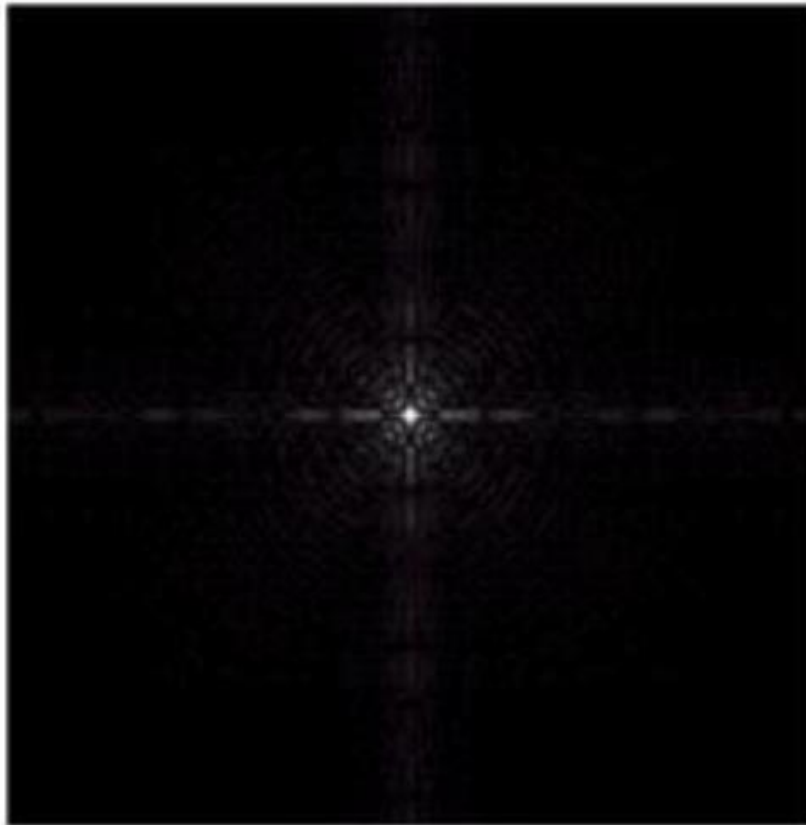


Original Image

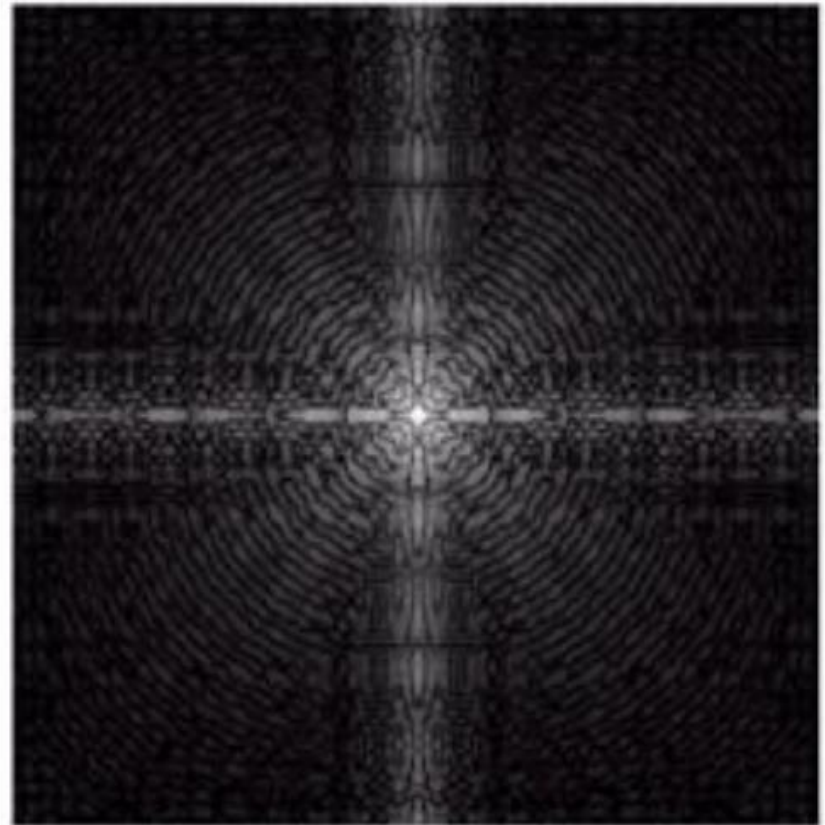


Mapped image

Example of Logarithm Image

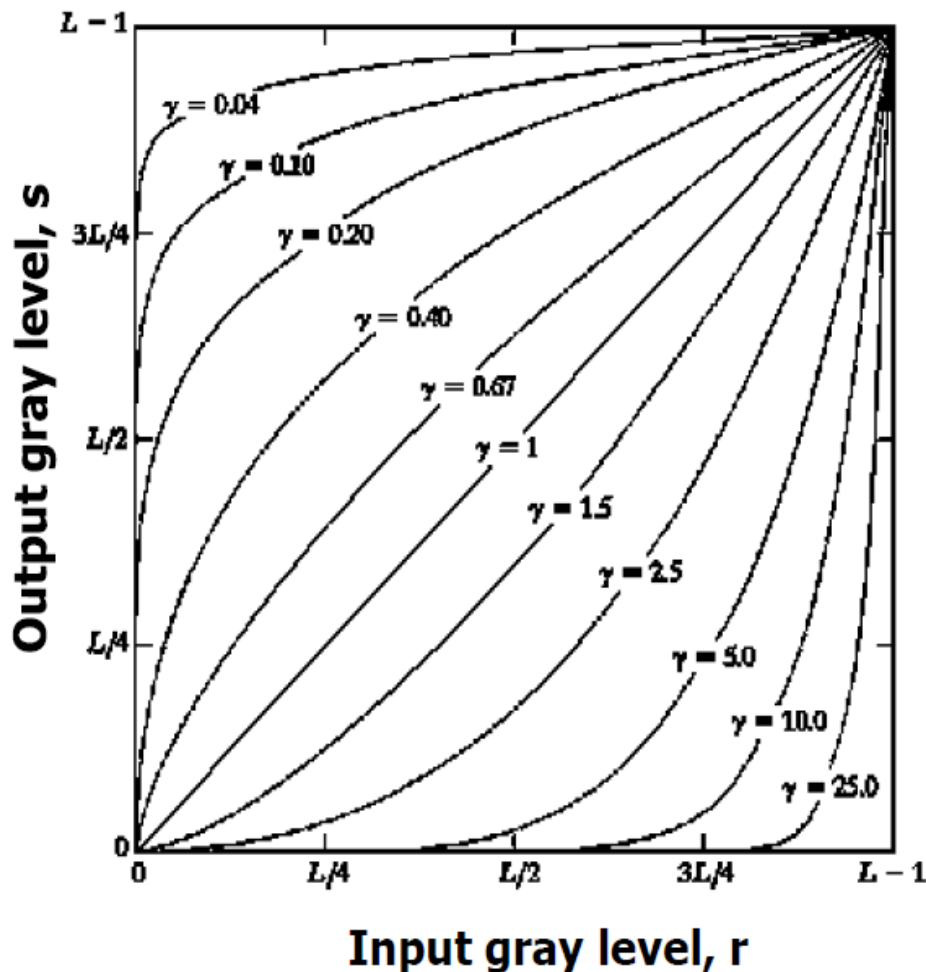


**Fourier Spectrum with
range = 0 to 1.5×10^6**



**Result after apply the log
transformation with $c = 1$,**

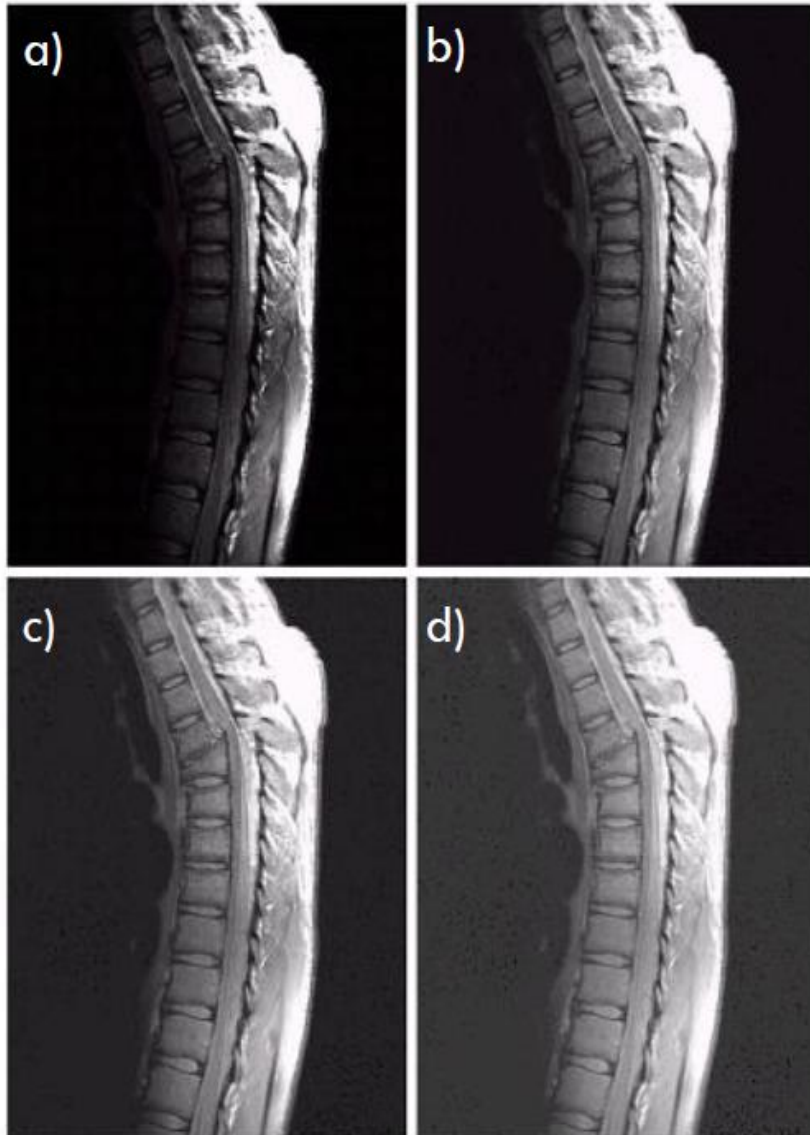
Power-Law Transformations



$$s = cr^\gamma$$

- c and γ are positive constants
- Power-law curves with fractional values of γ map a narrow range of dark input values into a wider range of output values, with the opposite being true for higher values of input levels.
- $c = \gamma = 1 \Rightarrow$ Identity function

Example : MRI



(a) a magnetic resonance image of an upper thoracic human spine with a fracture dislocation and spinal cord impingement

- ▣ The picture is predominately dark
- ▣ An expansion of gray levels are desirable $\Rightarrow$ needs $\gamma < 1$

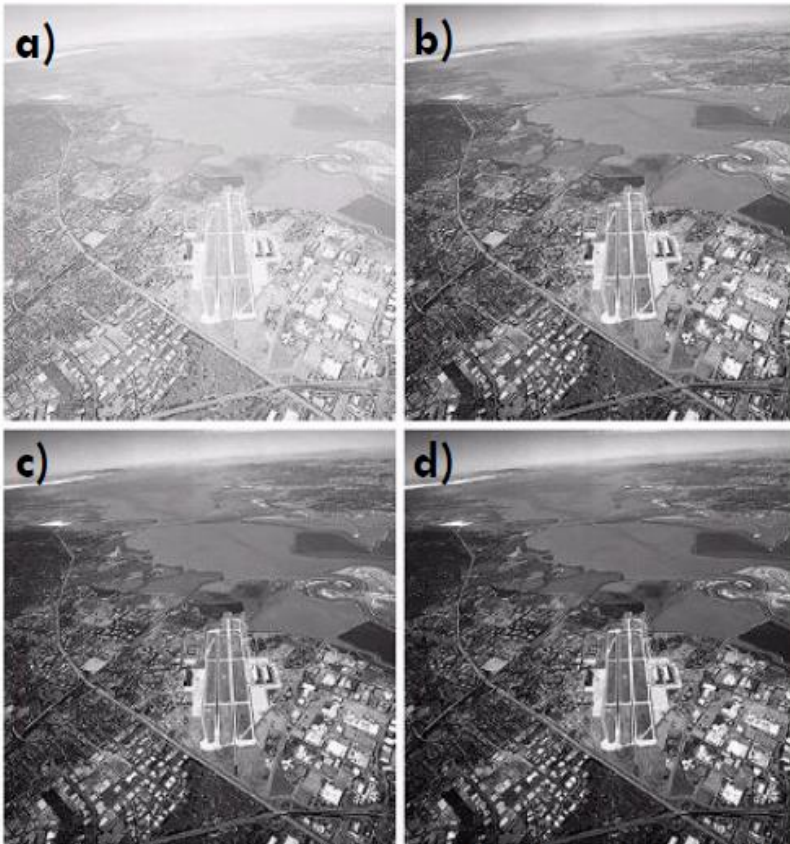
(b) result after power-law transformation with $\gamma = 0.6$, $c = 1$

(c) transformation with $\gamma = 0.4$
(best result)

(d) transformation with $\gamma = 0.3$
(under acceptable level)

Effect of decreasing gamma

- When the γ is reduced too much, the image begins to reduce contrast to the point where the image started to have very slight “wash-out” look, especially in the background



(a) image has a washed-out appearance, it needs a compression of gray levels $\Rightarrow$ needs $\gamma > 1$

(b) result after power-law transformation with $\gamma = 3.0$ (suitable)

(c) transformation with $\gamma = 4.0$ (suitable)

(d) transformation with $\gamma = 5.0$ (high contrast, the image has areas that are too dark, some detail is lost)

Piecewise-Linear Transformation Functions

- A complementary approach to the previous methods
- The form of piecewise functions can be arbitrarily complex
- Practical implementation of some important transformations can be formulated only as piecewise functions
- Their specification requires considerably more user input

Contrast Stretching

- The Simplest of piecewise linear functions.
- **Basic idea** – to increase the dynamic range of the gray levels in the image being processed.
- During image acquisition, low contrast images may result due to
 - Poor illumination
 - Lack of dynamic range in image sensor
 - Wrong setting of the lens aperture

Contrast Stretching

□ Contrast transform result

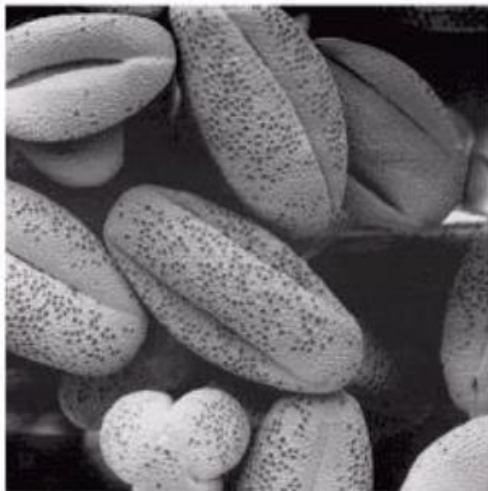
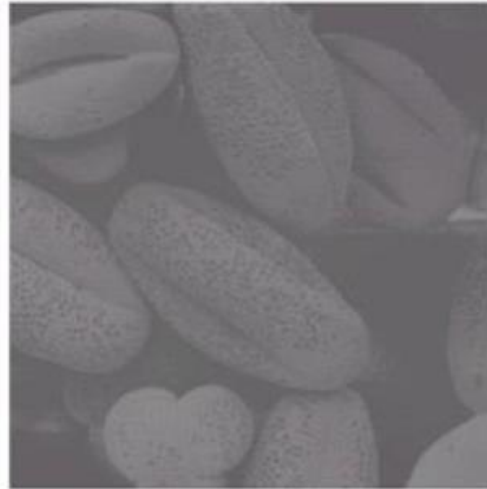
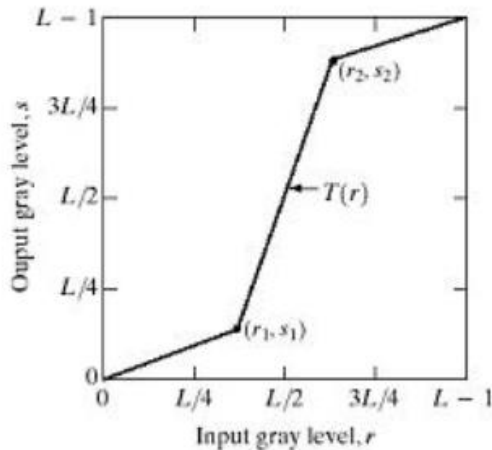


Original



Enhanced

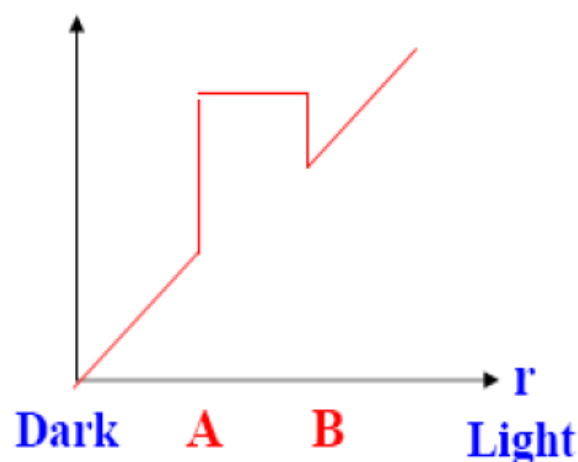
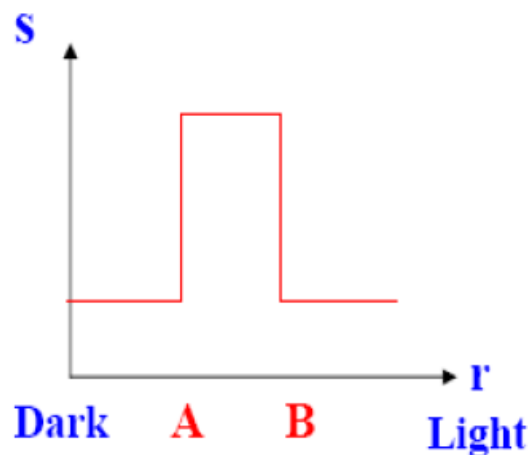
Contrast Stretching: Example



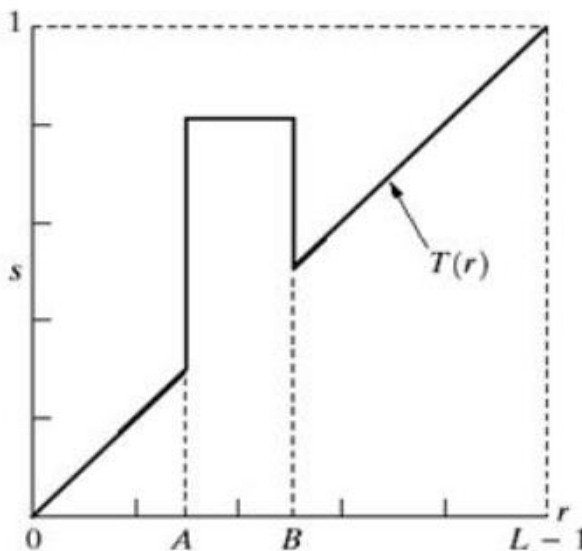
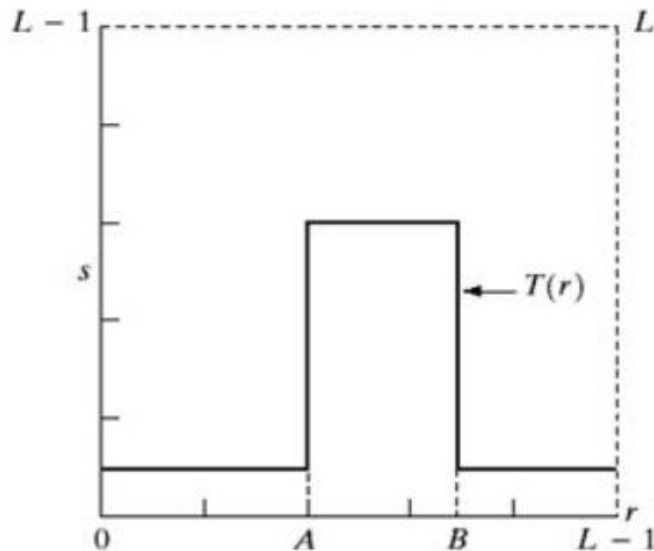
- increase the dynamic range of the gray levels in the image
- (b) a low-contrast image
- (c) result of contrast stretching: $(r_1, s_1) = (r_{\min}, 0)$ and $(r_2, s_2) = (r_{\max}, L-1)$
- (d) result of thresholding

Gray-level slicing

- Highlight a specific range of gray values
- Two basic Methods
 - ▣ Display a high value for all gray levels in the range of interest and a low value for all other
 - ▣ Brighten the desired range of gray levels but preserve all other levels

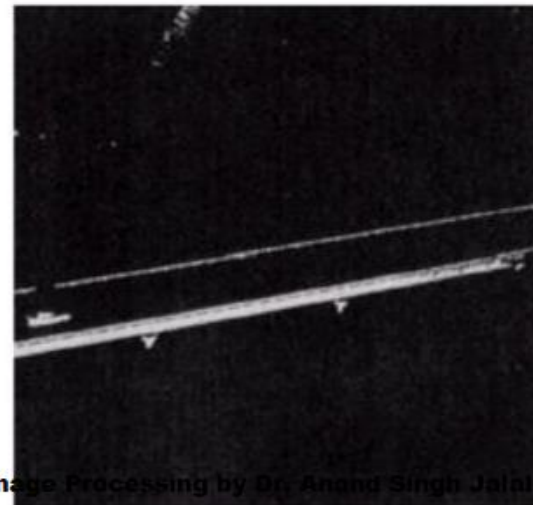


Gray-level slicing: Example

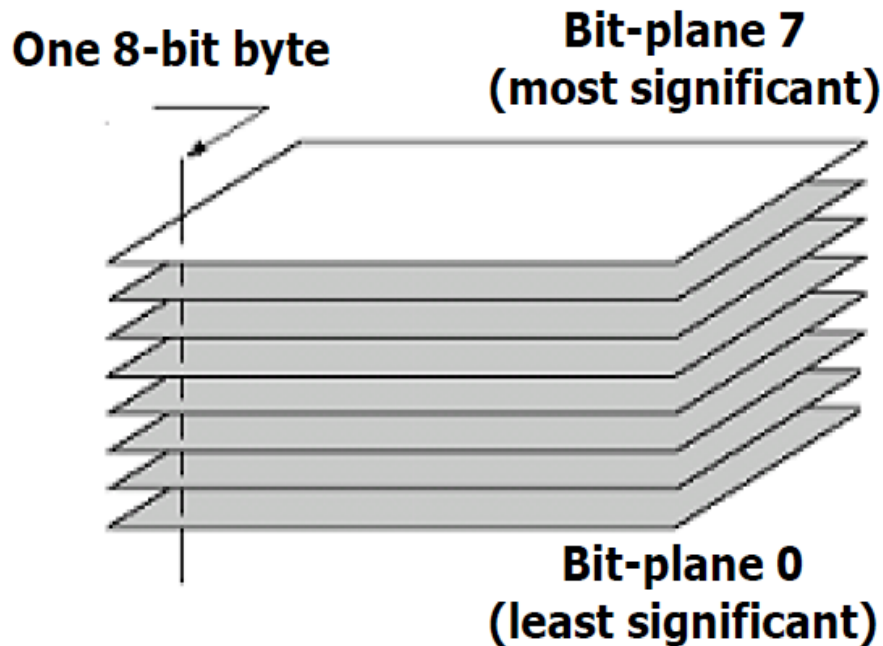


a	b
c	d

- (a) This transformation highlights range $[A, B]$ of gray levels and reduces all others to a constant level.
- (b) This transformation highlights range $[A, B]$ but preserves all other levels.
- (c) An image.
- (d) Result of using the transformation in (a).

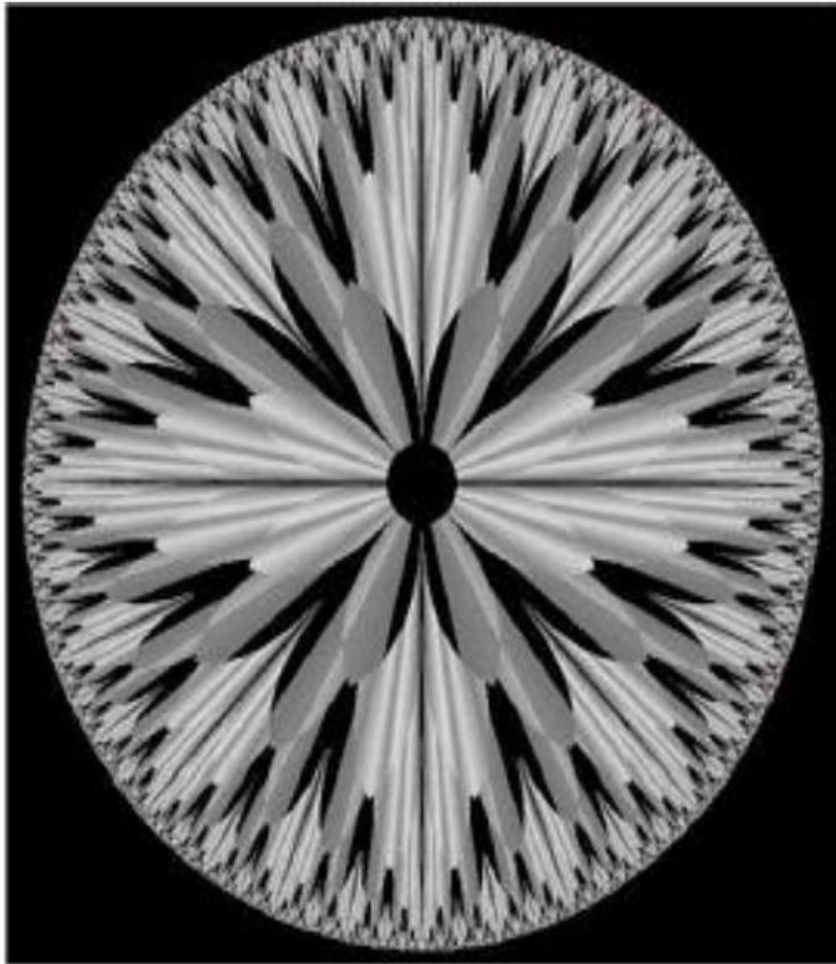


Bit-plane slicing



- Highlighting the contribution made to total image appearance by specific bits
- Suppose each pixel is represented by 8 bits
- Higher-order bits contain the majority of the visually significant data
- Useful for analyzing the relative importance played by each bit of the image

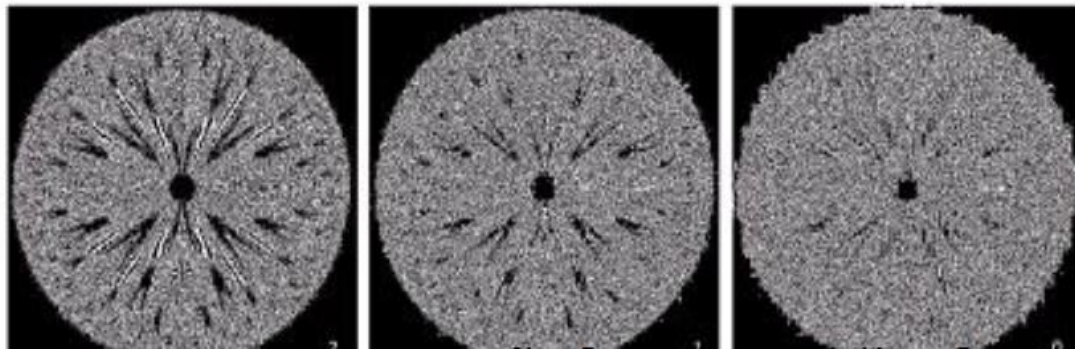
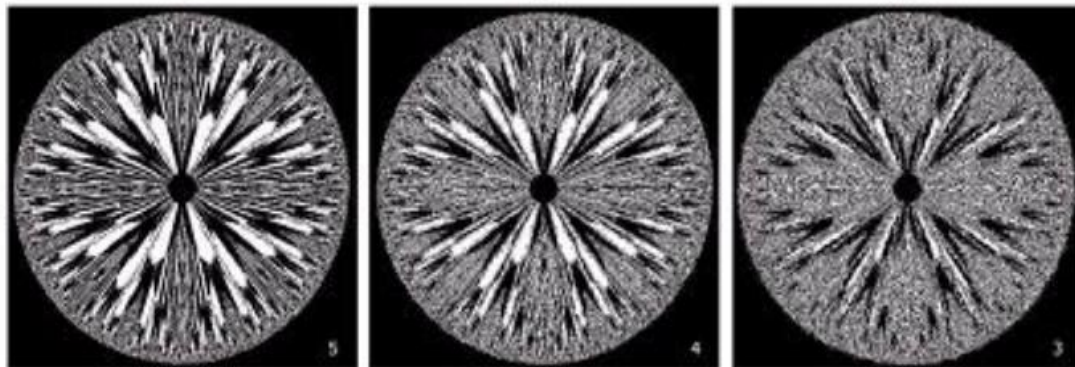
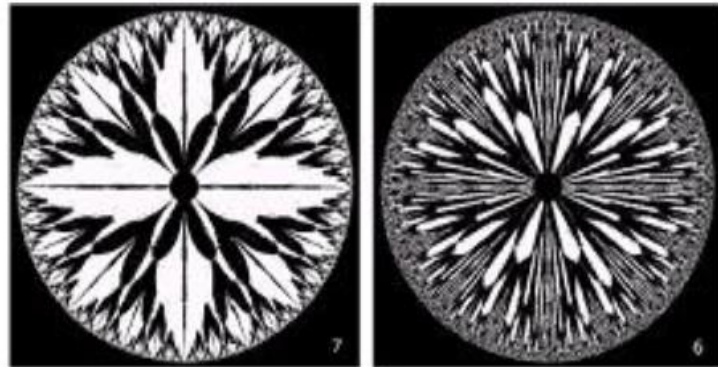
Bit-plane slicing: Example



An 8-bit fractal image

- Higher order bit planes of an image carry a significant amount of visually relevant details.
- Lower order planes contribute more to fine (often imperceptible) details.
- The (binary) image for bit-plane 7 can be obtained by processing the input image with a thresholding gray-level transformation.
 - Map all levels between 0 and 127 to 0
 - Map all levels between 129 and 255 to 255

Bit-plane slicing: Example (8 bit planes)



Bit-plane 7		Bit-plane 6	
Bit-plane 5	Bit-plane 4	Bit-plane 3	
Bit-plane 2	Bit-plane 1	Bit-plane 0	